

4

1. (a) Prove that if a is the only element of order 2 in a group G , then a must lie in the center of the group.

Proof. Suppose $a \in G$ and is the only element in G of order 2. To show that $a \in Z(G)$, we must show that $\forall t \in G, ta = at$. For any $x \in G$, consider the expression axa^{-1} . To show that $a = axa^{-1}$, we need to show that axa^{-1} is also of order 2. Squaring axa^{-1} ,

$$(axa^{-1})^2 = (axa^{-1}) \cdot (axa^{-1}).$$

Using the property of associativity from G ,

$$\begin{aligned} (axa^{-1})^2 &= x(ax^{-1}) \cdot (xa)x^{-1} \\ &= xa(x^{-1}) \cdot (x)ax^{-1} \\ &= xa(x^{-1} \cdot x)ax^{-1} \\ &= xaeax^{-1} \\ &= xaa^2ax^{-1} \\ &= xa^2a^2x^{-1} \\ &= xx^{-1} = e. \end{aligned}$$

Is 2 the smallest positive integer that satisfies this equality? Consider

$$axa^{-1} = e.$$

Multiplying x on the RHS and x^{-1} on the LHS of both expressions,

$$\begin{aligned} a &= x^{-1}(e)x \\ &= x^{-1}x \\ &= e. \end{aligned}$$

But this implies that $|a| = 1$, which contradicts the assumption of $|a| = 2$. So, it must be that $|axa^{-1}| = 2$. But, a is the only element in G of order 2. So,

$$a = axa^{-1}.$$

Multiplying x on the RHS of both expressions,

$$\begin{aligned} ax &= axa^{-1}x \\ \Rightarrow ax &= xa. \end{aligned}$$

So, $a \in Z(G)$. □

(b) Prove that no group can have exactly two elements of order 2.

Proof. By contradiction, let $a, b \in G$ s.t they're the only elements of order 2 in G . However, from (a), we know that $\forall c \in G$, we can use a and b to find more expressions of order 2 such as cac^{-1} or cbc^{-1} . $\therefore$ there can't be exactly two elements of order 2 in any group. □

2. Let G be a group, and let H be a subgroup of G . For any fixed x in G , define

$$xHx^{-1} := \{xhx^{-1} \mid h \in H\}$$

(a) Prove that xHx^{-1} is a subgroup of G .

Proof. Is xHx^{-1} nonempty? Since $H \leq G$, $e \in H$. So,

$$\begin{aligned} xex^{-1} &= xx^{-1} = e \\ \Rightarrow e &\in xHx^{-1}. \end{aligned}$$

So, xHx^{-1} is nonempty. By the One-Step Subgroup Test, let xHx^{-1} be a subset of G and choose any two arbitrary elements of the form $xa x^{-1}, xb x^{-1} \in xHx^{-1}$ where $a, b \in H$. To show that $xHx^{-1} \leq G$, we need to show that $(xa x^{-1})(xb x^{-1})^{-1} \in xHx^{-1}$. Using the Socks-Shoes Property,

$$(xbx^{-1})^{-1} = (x^{-1})^{-1}b^{-1}x^{-1} = xb^{-1}x^{-1}.$$

Applying the One-Step Subgroup Test,

$$\begin{aligned} (xa x^{-1})(xb^{-1}x^{-1}) &= x(ax^{-1})(xb^{-1})x^{-1} \\ &= xa(x^{-1})(x)b^{-1}x^{-1} \\ &= xa(x^{-1}x)b^{-1}x^{-1} \\ &= x(ab^{-1})x^{-1}. \end{aligned}$$

Since $H \leq G$ and $a, b \in H$, $ab^{-1} \in H$. So, $x(ab^{-1})x^{-1} \in xHx^{-1}$ and $xHx^{-1} \leq G$. □

(b) If H is cyclic, prove that xHx^{-1} is also cyclic.

Proof. Suppose H is cyclic. Then, $\exists$ some element $a \in H$ such that $\langle a \rangle = H$. To show that xHx^{-1} is cyclic, we want to show that $\exists$ an element that generates xHx^{-1} . That is, $\exists$ an element $xa x^{-1}$ where $a \in H$ s.t. $\langle xa x^{-1} \rangle = xHx^{-1}$. We know that $\forall h \in H$, we can rewrite it as

$$h = a^n \text{ for some } n \in \mathbb{Z}.$$

Consider squaring and cubing axx^{-1} ,

$$\begin{aligned}
 (axx^{-1})^2 &= (axx^{-1})(axx^{-1}) \\
 &= x(ax^{-1})(xa)x^{-1} \\
 &= xa(x^{-1})(x)ax^{-1} \\
 &= xa(x^{-1}x)ax^{-1} \\
 &= xaaax^{-1} \\
 &= xa^2x^{-1} \text{ where } a^2 \in H
 \end{aligned}$$

$$\begin{aligned}
 (axx^{-1})^3 &= (axx^{-1})(axx^{-1})(axx^{-1}) \\
 &= (xa^2x^{-1})(axx^{-1}) \\
 &= x(a^2x^{-1})(xa)x^{-1} \\
 &= xa^2(x^{-1})(x)ax^{-1} \\
 &= xa^2(x^{-1}x)ax^{-1} \\
 &= xa^2ax^{-1} \\
 &= xa^3x^{-1} \text{ where } a^3 \in H.
 \end{aligned}$$

In general, consider the $n - 1$ power of axx^{-1} ,

$$\begin{aligned}
 (axx^{-1})^{n-1} &= \underbrace{(xa_{n-1}x^{-1})(xa_{n-2}x^{-1})(xa_{n-3}x^{-1}) \dots (axx^{-1})}_{n-2 \text{ terms}} \\
 &= (xa^{n-2}x^{-1})(axx^{-1}) \\
 &= x(a^{n-2}x^{-1})(xa)x^{-1} \\
 &= xa^{n-2}(x^{-1})(x)ax^{-1} \\
 &= xa^{n-2}(x^{-1}x)ax^{-1} \\
 &= xa^{n-2}ax^{-1} \\
 &= xa^{n-1}x^{-1} \text{ where } a^{n-1} \in H.
 \end{aligned}$$

So,

$$\begin{aligned}
 \langle axx^{-1} \rangle &= \{axx^{-1}, (axx^{-1})^2, (axx^{-1})^3, \dots, (axx^{-1})^{n-1}, e\} \\
 &= xHx^{-1}.
 \end{aligned}$$

So, xHx^{-1} is cyclic. □

(c) If H is Abelian, prove that xHx^{-1} is also Abelian.

Proof. Suppose H is Abelian. Let $axx^{-1}, bxx^{-1} \in xHx^{-1}$ where $a, b \in H$. We want to show that $(axx^{-1})(bxx^{-1}) = (bxx^{-1})(axx^{-1})$. Taking the product of the LHS,

$$\begin{aligned}
 (axx^{-1})(bxx^{-1}) &= x(ax^{-1})(xb)x^{-1} \\
 &= xa(x^{-1})(x)bx^{-1} \\
 &= xa(x^{-1}x)bx^{-1} \\
 &= x(ab)x^{-1} \text{ where } ab \in H.
 \end{aligned}$$

But, since H is Abelian, $ab = ba \forall a, b \in H$. So,

$$\begin{aligned}(xax^{-1})(xbx^{-1}) &= x(ba)x^{-1} \\ &= xb(x^{-1}x)ax^{-1} \\ &= xb(x^{-1})(x)ax^{-1} \\ &= x(bx^{-1})(xa)x^{-1} \\ &= (xbx^{-1})(xax^{-1}).\end{aligned}$$

So, xHx^{-1} is Abelian. □

3. (a) In $G_1 = (\mathbb{Z}_{24}, +)$, list all generators for the subgroup of order 8. More generally, if $G_2 = \langle a \rangle$ with $|a| = 24$, list all the generators for the subgroups of order 8.

$$* \mathbb{Z}_n = \{0, 1, 2, \dots, n-1\}, \text{ Divisors of 24: } 1, 2, 3, 4, 6, 8, 12, 24$$

Divisors of 24

$$1) \boxed{\langle 3 \rangle = \{3, 6, 9, 12, 15, 18, 21, 0\}}$$

Non-Divisors of 24

$$2) \boxed{\langle 9 \rangle = \{9, 18, 3, 12, 21, 6, 15, 0\}}$$

$$3) \boxed{\langle 15 \rangle = \{15, 6, 21, 12, 3, 18, 9, 0\}}$$

$$4) \boxed{\langle 21 \rangle = \{21, 18, 15, 12, 9, 6, 3, 0\}}$$

More generally,

$$\boxed{\langle a^3 \rangle = \{a^3, a^6, a^9, a^{12}, a^{15}, a^{18}, a^{21}, e\}}$$

$$\boxed{\langle a^9 \rangle = \{a^9, a^{18}, a^3, a^{12}, a^{21}, a^6, a^{15}, e\}}$$

$$\boxed{\langle a^{15} \rangle = \{a^{15}, a^6, a^{21}, a^{12}, a^3, a^{18}, a^9, e\}}$$

$$\boxed{\langle a^{21} \rangle = \{a^{21}, a^{18}, a^{15}, a^{12}, a^9, a^6, a^3, e\}}$$

(b) Let a belong to a group with $|a| = 100$. Find $|a^{98}|$ and $|a^{70}|$.

$$* |a^k| = \left| \frac{n}{\gcd(n, k)} \right|$$

$$\Rightarrow |a^{98}| = \frac{100}{\gcd(100, 98)} = \frac{100}{2} = \boxed{50}$$

$$\Rightarrow |a^{70}| = \frac{100}{\gcd(100, 70)} = \frac{100}{10} = \boxed{10}$$

4. Give an example of a group with exactly 7 subgroups (including the trivial group and the group itself). Generalize to a group with exactly n subgroups for any positive integer n .

Consider the cyclic group $(\mathbb{Z}_{64}, +)$

- $\{e\} \rightarrow$ trivial group
- $\{1, 2, 3, \dots, 63, e\} \rightarrow$ entire group $(\langle 1 \rangle)$
- $\{2, 4, 6, 8, \dots, 62, e\} \rightarrow \langle 2 \rangle$
- $\{4, 8, 12, 16, \dots, 60, e\} \rightarrow \langle 4 \rangle$
- $\{8, 16, 24, 32, 40, 48, 56, e\} \rightarrow \langle 8 \rangle$
- $\{16, 32, 38, e\} \rightarrow \langle 16 \rangle$
- $\{32, e\} \rightarrow \langle 32 \rangle$

Note that $64 = 2^6$ and has $\underline{7}$ subgroups. If the order of a cyclic group is equal to a prime number raised to a power, the total divisors, or subgroups of the group, is equal to one less the power. In general, if a group G is of order k , it will have exactly n subgroups if

$$k = p^{n-1} \text{ where } p \text{ is prime}$$

for any $n \in \mathbb{Z}$.

5. (a) If H and K are subgroups of a group G , show that $H \cap K$ is also a subgroup of G .

Proof. Suppose H and K are subgroups of G . By the One-Step Subgroup Test, we need to show that $ab^{-1} \in H \cap K \forall a, b \in H \cap K$. For any $a, b \in H \cap K$, a and b are in both H and K . So, we have to show that ab^{-1} is in both H and K . But, since $H \leq G$ and $K \leq G$, they both satisfy the One-Step Subgroup Test. So, $\exists ab^{-1} \in H \forall a, b \in H$ and $\exists ab^{-1} \in K \forall a, b \in K$. So, $ab^{-1} \in H \cap K$. $\therefore H \cap K \leq G$. $\square$

(b) Let a and b belong to a group G . If $|a|$ and $|b|$ are finite and relatively prime, show that $\langle a \rangle \cap \langle b \rangle = \langle e \rangle$.

Proof. Suppose $a, b \in G$ and the orders of a and b are finite and relatively prime. Let $n = |a|$ and $m = |b|$. By Bezout's Identity, $\exists s, t \in \mathbb{Z}$ s.t

$$ns + mt = 1.$$

Let $x \in \langle a \rangle \cap \langle b \rangle$. Then, $x = a^t$ for some $t \in \mathbb{Z}$ and $x = b^h$ for some $h \in \mathbb{Z}$. But,

$$|a| = n \Rightarrow a^n = e$$

$$|b| = m \Rightarrow b^m = e.$$

So,

$$\begin{aligned} x^n &= (a^t)^n = a^{tn} = (a^n)^t = e^t = e \\ x^m &= (b^h)^m = b^{hm} = (b^m)^h = e^h = e. \end{aligned}$$

That means we can rewrite x as

$$\begin{aligned}x &= x^{ns+mt} \\&= x^{ns} \cdot x^{mt} \\&= (x^n)^s \cdot (x^m)^t \\&= e^s \cdot e^t \\&= e \\&= \{e\}.\end{aligned}$$

So, $\langle a \rangle \cap \langle b \rangle = \{e\}$.

□